AI CONSULTING CAPSTONE · ROUND 2

Assist

Complaint operations for a mid-size EU bank.
Three decisions proposed. Every one confirmed by a person.

Ugo Ahukannah
Financial services · mid-size EU retail bank

Complaint triage   Reporting assistance   Anomaly flagging

What changed after Round 1

TAKEAWAY Same industry, same client, same lead use case. The scope widened from one use case to three.

ROUND 1 — THE PITCH		ROUND 2 — WHAT I BUILT
Three use cases proposed, one built	→	**Three proposed, three built**
A workflow you could watch	→	**A working application you can use**
Cost estimate	→	**ROI, break-even and a commercial model**
"AI Act work deferred to Round 2"	→	**Classification, reasoning, conformity summary**

Decision: KEEP. Nothing challenged the sector or the honesty position, so nothing changed direction.

Three jobs, one shape

TAKEAWAY High volume, low complexity, unstructured text — the work a mid-size firm can neither automate with rules nor afford to staff.

EVERY DAY

Routing

2,426 complaints a year. 64 categories. Two handlers reading the same complaint pick different ones.

EVERY QUARTER

Reporting

The regulatory return is assembled by hand, under time pressure, by the people least able to spare it.

TOO LATE

Fraud

7.4% of complaints are unauthorised transactions. By the time one arrives, the money has gone.

The blocker isn't technical. A decision nobody can see is a decision nobody can sign off.

One decision spine, three capabilities

TAKEAWAY Every capability is the same four steps. Only the evidence changes.

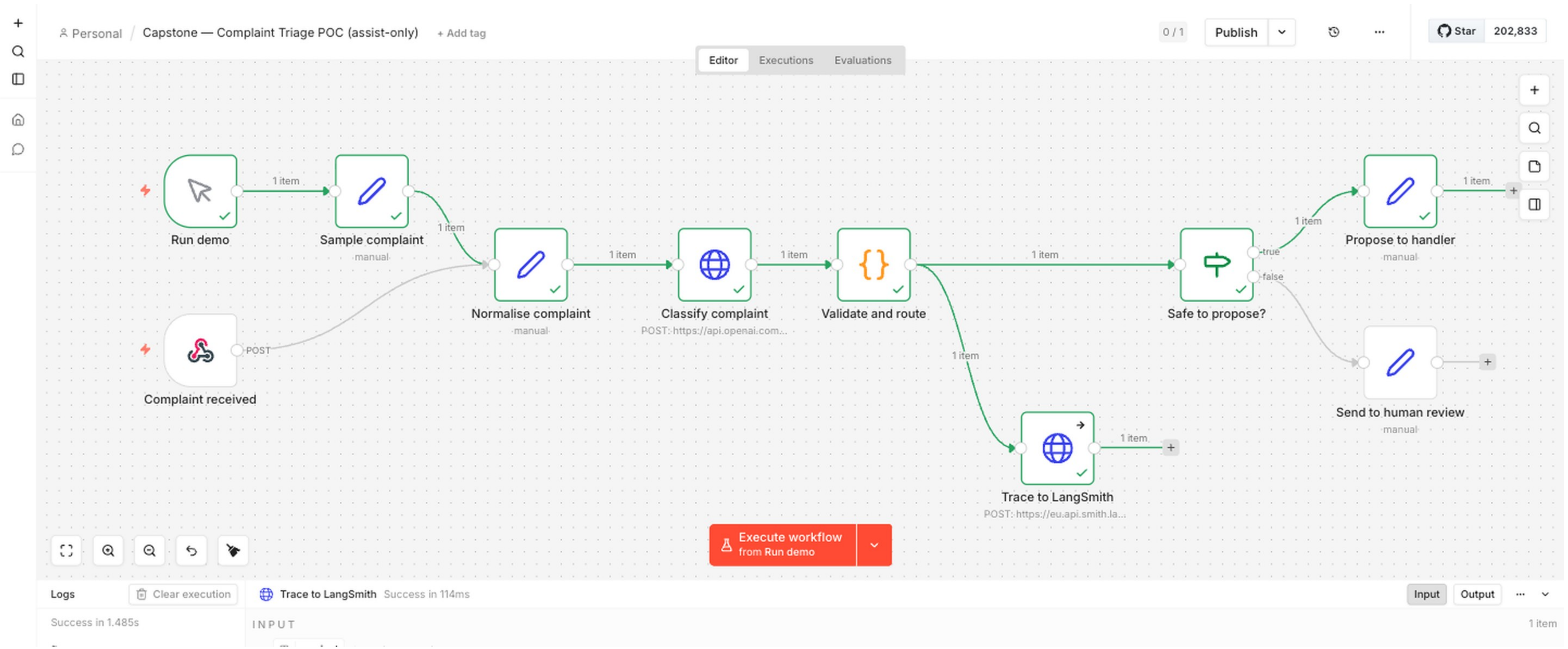
Validate the input → The model proposes → Guards check the evidence → A PERSON DECIDES

Capability	It proposes	Grounded in
Complaint triage	A routing queue	A verbatim quote from the complaint
Reporting assistance	Report prose	Figures this system computed
Anomaly flagging	An analyst case note	The transaction's own values

One batch, three capabilities, three kinds of fabrication caught: **2 invented quotes · 6 invented values · 1 uncomputed figure.**

The proof of concept

TAKEAWAY Nine steps, left to right. Green means it ran. It proposed a team, quoted your customer, and stopped.



€ What it returns

TAKEAWAY A bespoke build does not pay back for a firm your size. −31% over 36 months, break-even in month 69.

36-month ROI	What would have to change
−31.0%	As proposed — I build it for you, I run the oversight
−9.4%	Your team owns the quarterly review after handover
+16.9%	The build is paid for once and shared across five firms
+96.6%	Both

€18,200 buys a decision, not a deployment. The capability works. The commercial shape is what has to change.

The three that could derail this

TAKEAWAY The two biggest are not technical. They are about how it is deployed and measured.

Risk	L × I	What we do about it
The value assumptions are wrong	4 × 5	The pilot measures those two numbers. Fixed fee per phase — you can stop at €18,200
Handlers stop reading	4 × 4	Acceptance above 97% is a failure condition, not a success. Measured per handler
Oversight costs more than it saves	3 × 4	One spine, so one review covers all three. Instrumented from day one

Twelve risks scored in full — regulatory, technical, ethical, operational. These are the top three.

Where this sits in law

TAKEAWAY Not high-risk under the AI Act — and the reason matters more than the label.

EU AI ACT

- Not Annex III. It never touches creditworthiness
- Article 4 AI literacy applies today, whatever the tier
- Two questions for your counsel, both on anomaly flagging

GDPR

- You are the controller, and also the provider
- The narrative goes to the model in full — pseudonyms protect the reference, not the content
- Transfer work is a gate before any pilot on real data

What holds it out of the high-risk tier is a design property: **it proposes, a person decides.**

How this would go

TAKEAWAY Shadow mode first: it proposes alongside your team, nobody acts on it, and we measure the gap.

0 — Discovery
€5,600 · 4 weeks

Two of your handlers label 300 complaints. That ground truth is worth buying even if you never build this.

1 — Pilot
€12,600 · 12 weeks

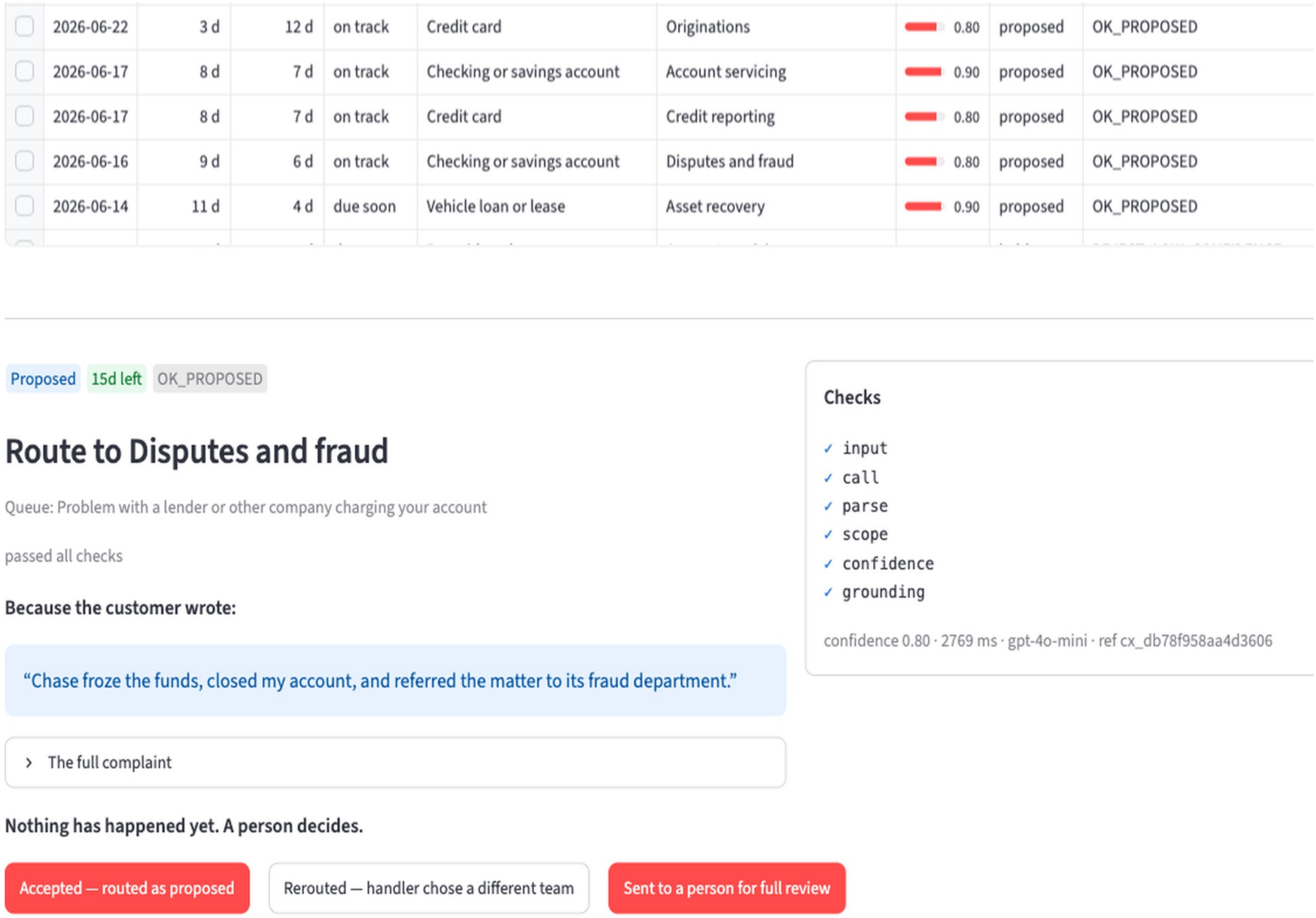
All three running in shadow for 60 days. Zero operational risk. Ends in a go / no-go.

2 — Deploy
€14,000 · 10 weeks

Only if accuracy ≥80%, acceptance 75–97%, and zero ungrounded outputs reached a handler.

Commercially: a licensed product implemented by consulting — not SaaS, not a bespoke build.

The beginning of the real product



What the POC only implied

- A queue, already read. Nobody types a complaint into a box it already arrived in.
- A deadline column — the question your regulator actually asks.
- Every check it ran, by name, and the one it failed.
- Nothing happens until someone presses a button — and the press is recorded beside the proposal.

What I'm asking for

A 60-day shadow pilot on your own complaints.

€18,200 to Phase 1. Nobody acts on a single suggestion.

It works — three capabilities running, every decision checked and traced.

It is lawful — not high-risk, with the reasoning written down and two questions named for your counsel.

It does not yet pay for itself at your size — and I would rather you heard that from me now than found it in month three.

Backup

Backup · Why 60.5% is not accuracy

TAKEAWAY The model is consistent. It disagrees with an untrained label.

Measure	Value	What it means
Agreement with CFPB labels	60.5%	Those labels were picked by the complainant from a dropdown
Self-agreement, same input twice	88.0%	It is consistent — so a cleverer model is not the fix
Fabricated quotes caught	4 of 60	Confidently invented. Every one stopped by the guard

Phase 0 creates the expert labels that would make an accuracy claim possible.

Backup · The anomaly detector's 100%

TAKEAWAY It finds 100% of planted anomalies. Do not report that as a detection rate.

WHAT IT LEGITIMATELY SHOWS

- Deterministic — same batch, same candidates, every time
- Covers all four patterns, not just the easy one
- 97.1% precision on 2,069 transactions

WHY IT IS NOT ACCURACY

- The thresholds and the data generator were written by the same author
- So it measures whether the code agrees with itself
- Same trap as the 60.5%. Only a pilot on real traffic settles it

The caveat travels with the figure in the code, so it cannot be lifted onto a slide alone.

Backup · AI Act classification, step by step

TAKEAWAY Six steps. The one that matters is step 4.

STEP	FINDING
1 · Is it an AI system?	Yes, all three
2 · Prohibited under Art. 5?	No — one close call on anomaly flagging, flagged for counsel
3 · Annex I safety component?	No
4 · An Annex III use case?	No. Never evaluates creditworthiness — Annex III(5)(b) does not bite
5 · Art. 6(3) derogation	Not reached. If step 4 fails, anomaly flagging alone becomes high-risk — it profiles
6 · Art. 50 transparency	No customer-facing surface. Generated drafts marked anyway

Backup · What actually leaves the building

TAKEAWAY Pseudonymisation protects the reference. It does not protect the content.

	Identifier	Narrative content
Sent to the model provider (US)	Never	In full
Stored in monitoring (EU)	Never	The quoted sentence
In the decision log	Never	The quoted sentence

So the transfer work — DPA, transfer impact assessment, zero retention — is a gate before any pilot on real data, not a Phase 2 discovery.

Backup · What the pilot has to prove

TAKEAWAY Miss any of the first three and we do not proceed.

GATE	THRESHOLD
Team-level accuracy vs expert labels	≥ 80%
Handler acceptance rate	≥ 75% AND ≤ 97%
Ungrounded outputs reaching a handler	Zero. Absolute
Correct abstention	≥ 95%
Anomaly precision on analyst disposition	≥ 30% — if this alone fails, ship the other two

Backup · Anomaly flagging

☐	2026-06-20	2 d	threshold_structuring	150.7×	€4935.00	5	held	REJECT_VALUE_MISMATCH
☐	2026-06-20	2 d	amount_spike	54.8×	€995.00	1	proposed	OK_PROPOSED

Proposed OK_PROPOSED

Amount spike

a single amount far above this account's own normal

Transactions	Total	Account median	Times normal
1	€990	€18	54.5×

Case note:

This candidate was raised due to a significant spike in the transaction amount, with a single transfer of EUR 990.0, which is far above the account's median transaction of EUR 18.15. The amount is approximately 54.5 times higher than what is typical for this account, indicating an unusual activity pattern.

Fastest check: Review the account's transaction history for any similar high-value transfers.

Nothing has happened yet. A person decides.

Escalated to an analyst | Dismissed — ordinary for this account | Needs more information

Checks

- ✓ `input`
- ✓ `call`
- ✓ `parse`
- ✓ `scope`
- ✓ `confidence`
- ✓ `grounding`

confidence 0.80 · 1377 ms · gpt-4o-mini · ref cx_65b1bc11dceaef3b

transfer · online_banking · IE

The division of labour

- A deterministic detector decides WHAT is unusual.
- The model only explains WHY, in language an analyst can act on.
- Ranked by departure from that account's own normal — never by the amount.
- Six of 56 case notes were stopped for citing a figure that was not in the record.